# TabuLLM: AI Benchmark Report

## 1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

| Audit Indicator | Global Session Mean (± Std) |
|---|---|
| Synthetic Fidelity (JSD) | 69.4 ± 4.98 |
| Correlation Stability | 90.1 ± 0.94 |
| Distribution Consistency | 77.8 ± 4.85 |
| Entropy Preservation | 63.3 ± 3.80 |
| Mean Z Score Drift | 0.3 ± 0.03 |
| Categorical Fidelity | 0.0 ± 0.00 |
| Privacy Risk | 0.0 ± 0.00 |
| Data Density | 100.0 ± 0.00 |
| Coverage Score | 98.4 ± 0.66 |
| Feature Accuracy | 65.2 ± 3.09 |
| Novelty Score | 0.8 ± 0.02 |

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## Dataset Audit: heart_failure_clinical_records.csv

### 1. Dataset Portrait

| Attribute | Value |
|---|---|
| Rows | 299 |
| Columns | 13 |
| Numeric features | 13 |
| Categorical features | 0 |
| Missing cells (%) | 0.0% |

### 2. Generation Strategy & LLM Config [1]

| Parameter | Configuration |
|---|---|
| Algorithm (*12) | google/gemma-4-e2b |
| Folds (*13) | 10 |
| Context % (*14) | 0% |
| Generation % (*15) | 100% |
| Join context (*16) | False |

### 3. Statistical Significance & Data Drift

| Feature | Orig Mean | Aug Mean | Shift | P-Value | Status |
|---|---|---|---|---|---|
| age | 60.834 | 63.365 | +2.531 | 4.2% | OK |
| anaemia | 0.431 | 0.339 | -0.093 | 21.4% | Warn |
| creatinine_phosphokinase | 581.839 | 191.172 | -390.668 | 67.1% | Drift |
| diabetes | 0.418 | 0.276 | -0.142 | 33.9% | Drift |
| ejection_fraction | 38.084 | 37.971 | -0.113 | 0.3% | OK |
| high_blood_pressure | 0.351 | 0.172 | -0.180 | 51.1% | Drift |
| platelets | 263358.029 | 254556.849 | -8801.180 | 3.3% | OK |
| serum_creatinine | 1.394 | 1.046 | -0.348 | 25.0% | Warn |
| serum_sodium | 136.625 | 137.310 | +0.684 | 0.5% | OK |
| sex | 0.649 | 0.812 | +0.163 | 25.1% | Drift |
| smoking | 0.321 | 0.038 | -0.283 | 88.3% | Drift |
| time | 130.261 | 163.439 | +33.178 | 25.5% | Drift |

### 4. Quality & Fidelity Metrics (Mean ± Std)

| Metric Category | Aggregated Score |
|---|---|
| Synthetic Fidelity (JSD) (*1) | 69.4 ± 4.98 |
| Correlation Stability (*2) | 90.1 ± 0.94 |
| Distribution Consistency (*3) | 77.8 ± 4.85 |
| Entropy Preservation (*4) | 63.3 ± 3.80 |
| Mean Z Score Drift (*5) | 0.3 ± 0.03 |
| Categorical Fidelity (*6) | 0.0 ± 0.00 |
| Privacy Risk (*7) | 0.0 ± 0.00 |

| | |
|---|---|
| Data Density (*8) | 100.0 ± 0.00 |
| Coverage Score (*9) | 98.4 ± 0.66 |
| Feature Accuracy (*10) | 65.2 ± 3.09 |
| Novelty Score (*11) | 0.8 ± 0.02 |

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

*[6] Synthetic Fidelity (JSD): Overall distribution similarity.*
*[7] Correlation Stability: Multivariate relationships preserved.*
*[8] Distribution Consistency: Median feature similarity.*
*[9] Entropy Preservation: Information diversity maintained.*
*[10] Mean Z Score Drift: Standardized distribution shift.*
*[11] Categorical Fidelity: Category distribution similarity.*
*[12] Privacy Risk: Synthetic rows too close to originals.*
*[13] Data Density: Non-missing cells in augmented data.*
*[14] Coverage Score: Synthetic coverage of original space.*
*[15] Feature Accuracy: Mean feature similarity.*
*[16] Novelty Score: Distance ratio to nearest neighbors.*

## 6. Downstream ML Utility: Comparative Analysis

### Model: AB

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.825 | 0.755 | 0.8233333333333335 | -0.00166666666666667 | |
| F1 | 0.7417404771442427 | 0.5253418653418654 | 0.7308240754611217 | -0.01091640168312096 | |

### Model: RF

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.8366666666666667 | 0.7433333333333334 | 0.8183333333333334 | -0.01833333333333335 | |
| F1 | 0.7454897762794181 | 0.4937121948143817 | 0.7059466952272475 | -0.03954308105217076 | |

### Model: GB

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.8316666666666667 | 0.7433333333333333 | 0.8283333333333334 | -0.00333333333333335 | |
| F1 | 0.7556715392060958 | 0.547697002410822 | 0.7396823313140702 | -0.01598920789202535 | |

### Model: MLP

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.5316666666666666 | 0.4933333333333333 | 0.5516666666666666 | 0.01999999999999999 | |
| F1 | 0.27138048447451885 | 0.20428087908444112 | 0.2475949585032719 | -0.02378552597124691 | |

### Model: LR

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.8066666666666666 | 0.7216666666666667 | 0.7766666666666666 | -0.02999999999999998 | |
| F1 | 0.6937863401621888 | 0.4133140604505113 | 0.5932719221343777 | -0.10051441802781107 | * |

### Model: SVM

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| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.6333333333333334 | 0.6333333333333334 | 0.6333333333333334 | 0 | |
| F1 | N/A | 0.0 | N/A | 0 | |

### Model: XGB

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.835 | 0.75 | 0.8366666666666667 | 0.00166666666666683 | |
| F1 | 0.7547418227770806 | 0.5306693394178759 | 0.752256924528976 | -0.00248489824810461 | |

### Model: NB

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.7499999999999999 | 0.7150000000000001 | 0.7066666666666667 | -0.04333333333333337 | |
| F1 | 0.5667171896628354 | 0.4349570738266156 | 0.4646967158182267 | -0.10202047384460858 | |

### Model: DT

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.7883333333333333 | 0.7233333333333333 | 0.7716666666666666 | -0.01666666666666663 | |
| F1 | 0.675837074942099 | 0.5808730799624362 | 0.6381922570640175 | -0.03764481787808155 | |

### Model: KNN

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.6133333333333335 | 0.6333333333333334 | 0.6333333333333334 | 0.01999999999999983 | |
| F1 | 0.007142857142857143 | 0.016 | 0.045059288537549404 | 0.037916431394692265 | |

## Appendix: Benchmark Data Trace

*Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline*

### A. Baseline: Original Dataset (Full Record Sample)

| # | age | anaemia | creatinine_phosphoki | diabetes | ejection_fractio | high_blood_pressu | platelets | serum_creatinine | serum_sodium | sex |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 75.0 | 0.0 | 582.0 | 0.0 | 20.0 | 1.0 | 265000.0 | 1.9 | 130.0 | 1.0 |
| 2 | 55.0 | 0.0 | 7861.0 | 0.0 | 38.0 | 0.0 | 263358.03 | 1.1 | 136.0 | 1.0 |
| 3 | 65.0 | 0.0 | 146.0 | 0.0 | 20.0 | 0.0 | 162000.0 | 1.3 | 129.0 | 1.0 |
| 4 | 50.0 | 1.0 | 111.0 | 0.0 | 20.0 | 0.0 | 210000.0 | 1.9 | 137.0 | 1.0 |
| 5 | 65.0 | 1.0 | 160.0 | 1.0 | 20.0 | 0.0 | 327000.0 | 2.7 | 116.0 | 0.0 |
| 6 | 90.0 | 1.0 | 47.0 | 0.0 | 40.0 | 1.0 | 204000.0 | 2.1 | 132.0 | 1.0 |
| 7 | 75.0 | 1.0 | 246.0 | 0.0 | 15.0 | 0.0 | 127000.0 | 1.2 | 137.0 | 1.0 |
| 8 | 60.0 | 1.0 | 315.0 | 1.0 | 60.0 | 0.0 | 454000.0 | 1.1 | 131.0 | 1.0 |
| 9 | 65.0 | 0.0 | 157.0 | 0.0 | 65.0 | 0.0 | 263358.03 | 1.5 | 138.0 | 0.0 |
| 10 | 80.0 | 1.0 | 123.0 | 0.0 | 35.0 | 1.0 | 388000.0 | 9.4 | 133.0 | 1.0 |
| 11 | 75.0 | 1.0 | 81.0 | 0.0 | 38.0 | 1.0 | 368000.0 | 4.0 | 131.0 | 1.0 |
| 12 | 62.0 | 0.0 | 231.0 | 0.0 | 25.0 | 1.0 | 253000.0 | 0.9 | 140.0 | 1.0 |
| 13 | 45.0 | 1.0 | 981.0 | 0.0 | 30.0 | 0.0 | 136000.0 | 1.1 | 137.0 | 1.0 |
| 14 | 50.0 | 1.0 | 168.0 | 0.0 | 38.0 | 1.0 | 276000.0 | 1.1 | 137.0 | 1.0 |
| 15 | 49.0 | 1.0 | 80.0 | 0.0 | 30.0 | 1.0 | 427000.0 | 1.0 | 138.0 | 0.0 |
| 16 | 82.0 | 1.0 | 379.0 | 0.0 | 50.0 | 0.0 | 47000.0 | 1.3 | 136.0 | 1.0 |
| 17 | 87.0 | 1.0 | 149.0 | 0.0 | 38.0 | 0.0 | 262000.0 | 0.9 | 140.0 | 1.0 |
| 18 | 45.0 | 0.0 | 582.0 | 0.0 | 14.0 | 0.0 | 166000.0 | 0.8 | 127.0 | 1.0 |
| 19 | 70.0 | 1.0 | 125.0 | 0.0 | 25.0 | 1.0 | 237000.0 | 1.0 | 140.0 | 0.0 |
| 20 | 48.0 | 1.0 | 582.0 | 1.0 | 55.0 | 0.0 | 87000.0 | 1.9 | 121.0 | 0.0 |

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## B.1 Audit Trace: Fold 1 | Gen: 100%

| # | age | anaemia | creatinine_phosphoki | diabetes | ejection_fractio | high_blood_pressu | platelets | serum_creatinine | serum_sodium | sex |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 62.859982544284094 | 0.0 | 222.0 | 0.0 | 12.0 | 0.0 | 145941.2430610823 | 0.9571489765270995 | 140.0 | 1.0 |
| 2 | 52.808220055265059 | 1.0 | 117.0 | 1.0 | 26.0 | 1.0 | 236402.69329390005 | 0.8890809679612339 | 139.0 | 0.0 |
| 3 | 55.676348205364688 | 0.0 | 158.0 | 0.0 | 28.0 | 0.0 | 141031.44914964645 | 1.1566407739656108 | 136.0 | 1.0 |
| 4 | 62.190992339018677 | 0.0 | 219.0 | 0.0 | 34.0 | 0.0 | 228194.05549649158 | 0.8897028997064366 | 135.0 | 1.0 |
| 5 | 62.493637099594366 | 0.0 | 124.0 | 0.0 | 40.0 | 0.0 | 297524.05647568416 | 0.7710460075487622 | 139.0 | 1.0 |
| 6 | 64.3933318602094 | 0.0 | 134.0 | 1.0 | 22.0 | 0.0 | 351804.50174372375 | 0.9022511549537533 | 136.0 | 1.0 |
| 7 | 45.0 | 1.0 | 66.0 | 1.0 | 25.0 | 0.0 | 233000.0 | 0.8 | 135.0 | 1.0 |
| 8 | 70.0 | 0.0 | 161.0 | 0.0 | 25.0 | 0.0 | 244000.0 | 1.2 | 142.0 | 0.0 |
| 9 | 58.0 | 1.0 | 400.0 | 0.0 | 40.0 | 0.0 | 164000.0 | 1.0 | 139.0 | 0.0 |
| 10 | 55.0 | 0.0 | 572.0 | 1.0 | 35.0 | 0.0 | 231000.0 | 0.8 | 143.0 | 0.0 |

## B.2 Audit Trace: Fold 2 | Gen: 100%

| # | age | anaemia | creatinine_phosphoki | diabetes | ejection_fractio | high_blood_pressu | platelets | serum_creatinine | serum_sodium | sex |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 49.715134866456344 | 0.0 | 198.0 | 0.0 | 41.0 | 0.0 | 164061.13958963082 | 1.0379562952358081 | 135.0 | 1.0 |
| 2 | 64.373423453068244 | 0.0 | 234.0 | 0.0 | 34.0 | 0.0 | 220522.11918563122 | 0.7234254978583747 | 136.0 | 1.0 |
| 3 | 55.047036762328474 | 0.0 | 29.0 | 0.0 | 37.0 | 0.0 | 134687.9485172542 | 0.9347121984117547 | 140.0 | 1.0 |
| 4 | 41.732444084795311 | 0.0 | 325.0 | 0.0 | 31.0 | 0.0 | 451000.17282106227 | 0.7995908259317095 | 137.0 | 1.0 |
| 5 | 48.639948937030211 | 1.0 | 200.0 | 0.0 | 24.0 | 0.0 | 320355.80194780476 | 1.0167835618035426 | 136.0 | 1.0 |
| 6 | 54.486647635223111 | 0.0 | 242.0 | 0.0 | 39.0 | 0.0 | 355226.91028747015 | 1.3700461326111646 | 140.0 | 1.0 |
| 7 | 60.0 | 0.0 | 2656.0 | 1.0 | 30.0 | 0.0 | 305000.0 | 2.3 | 137.0 | 1.0 |
| 8 | 45.0 | 1.0 | 130.0 | 0.0 | 35.0 | 0.0 | 174000.0 | 0.8 | 139.0 | 1.0 |
| 9 | 45.0 | 0.0 | 582.0 | 1.0 | 55.0 | 0.0 | 543000.0 | 1.0 | 132.0 | 0.0 |
| 10 | 65.0 | 1.0 | 59.0 | 1.0 | 60.0 | 0.0 | 172000.0 | 0.9 | 137.0 | 0.0 |

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## B.3 Audit Trace: Fold 3 | Gen: 100%

| # | age | anaemia | creatinine_phosphoki | diabetes | ejection_fractio | high_blood_pressu | platelets | serum_creatinine | serum_sodium | sex |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 48.628681713791075 | 0.0 | 196.0 | 0.0 | 45.0 | 0.0 | 281482.19404257287 | 0.8789977974641894 | 139.0 | 1.0 |
| 2 | 69.15303796534764 | 0.0 | 253.0 | 0.0 | 26.0 | 0.0 | 156915.4404971724 | 1.2509124263780622 | 138.0 | 1.0 |
| 3 | 65.4761991086699 | 0.0 | 62.0 | 0.0 | 37.0 | 0.0 | 291487.4849154005 | 0.8038165074274695 | 140.0 | 1.0 |
| 4 | 67.75563060569156 | 1.0 | 271.0 | 0.0 | 37.0 | 0.0 | 279290.16472121974 | 0.9018601008316104 | 141.0 | 1.0 |
| 5 | 65.31236492026586 | 1.0 | 196.0 | 0.0 | 32.0 | 0.0 | 292723.0429386996 | 1.1380706468709979 | 130.0 | 1.0 |
| 6 | 47.952081152791955 | 0.0 | 765.0 | 0.0 | 34.0 | 0.0 | 191408.39771318712 | 0.9294757409685358 | 142.0 | 1.0 |
| 7 | 50.0 | 0.0 | 582.0 | 1.0 | 38.0 | 0.0 | 310000.0 | 1.9 | 135.0 | 1.0 |
| 8 | 79.0 | 1.0 | 55.0 | 0.0 | 50.0 | 1.0 | 172000.0 | 1.8 | 133.0 | 1.0 |
| 9 | 65.0 | 1.0 | 160.0 | 1.0 | 20.0 | 0.0 | 327000.0 | 2.7 | 116.0 | 0.0 |
| 10 | 70.0 | 0.0 | 92.0 | 0.0 | 60.0 | 1.0 | 317000.0 | 0.8 | 140.0 | 0.0 |

## B.4 Audit Trace: Fold 4 | Gen: 100%

| # | age | anaemia | creatinine_phosphoki | diabetes | ejection_fractio | high_blood_pressu | platelets | serum_creatinine | serum_sodium | sex |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 70.63792206013885 | 0.0 | 300.0 | 0.0 | 29.0 | 0.0 | 213016.54422590256 | 0.9370306939906867 | 134.0 | 1.0 |
| 2 | 77.13640380236718 | 0.0 | 286.0 | 0.0 | 34.0 | 0.0 | 306493.6908337882 | 1.3166308610765243 | 141.0 | 1.0 |
| 3 | 72.1273385715185 | 0.0 | 95.0 | 0.0 | 33.0 | 0.0 | 265029.353854892 | 1.2316547164600717 | 139.0 | 1.0 |
| 4 | 60.153742520619446 | 1.0 | 186.0 | 0.0 | 26.0 | 1.0 | 239612.1604450699 | 0.7120370914841233 | 137.0 | 0.0 |
| 5 | 69.28927230715333 | 0.0 | 201.0 | 0.0 | 38.0 | 0.0 | 182729.66998795184 | 0.8721805572093032 | 139.0 | 1.0 |
| 6 | 57.21828299750444 6 | 1.0 | 103.0 | 0.0 | 40.0 | 0.0 | 314339.039917338 | 1.0949924698199667 | 148.0 | 1.0 |
| 7 | 48.0 | 1.0 | 131.0 | 1.0 | 30.0 | 1.0 | 244000.0 | 1.6 | 130.0 | 0.0 |
| 8 | 95.0 | 1.0 | 371.0 | 0.0 | 30.0 | 0.0 | 461000.0 | 2.0 | 132.0 | 1.0 |
| 9 | 45.0 | 1.0 | 981.0 | 0.0 | 30.0 | 0.0 | 136000.0 | 1.1 | 137.0 | 1.0 |
| 10 | 50.0 | 0.0 | 582.0 | 1.0 | 38.0 | 0.0 | 310000.0 | 1.9 | 135.0 | 1.0 |

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## B.5 Audit Trace: Fold 5 | Gen: 100%

| # | age | anaemia | creatinine_phosphoki | diabetes | ejection_fractio | high_blood_pressu | platelets | serum_creatinine | serum_sodium | sex |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 56.7366824225543 1 | 0.0 | 65.0 | 0.0 | 42.0 | 0.0 | 277613.578173786 7 | 1.10204963732986 5 | 139.0 | 0.0 |
| 2 | 73.3674860569028 2 | 0.0 | 152.0 | 0.0 | 37.0 | 0.0 | 229856.850206390 8 | 1.12212074526268 96 | 132.0 | 1.0 |
| 3 | 52.0597431180649 7 | 0.0 | 219.0 | 1.0 | 38.0 | 0.0 | 284538.322524980 06 | 1.06150981579857 14 | 139.0 | 1.0 |
| 4 | 56.8231788445204 84 | 0.0 | 199.0 | 0.0 | 33.0 | 0.0 | 248867.218330396 34 | 0.96267133813752 99 | 136.0 | 1.0 |
| 5 | 61.6384080066433 4 | 1.0 | 111.0 | 0.0 | 58.0 | 0.0 | 281939.528872382 7 | 0.86533642387941 39 | 132.0 | 0.0 |
| 6 | 80.5572879068848 8 | 0.0 | 165.0 | 1.0 | 29.0 | 0.0 | 256883.943623431 38 | 1.04052836935296 3 | 133.0 | 1.0 |
| 7 | 78.0 | 0.0 | 224.0 | 0.0 | 50.0 | 0.0 | 481000.0 | 1.4 | 138.0 | 1.0 |
| 8 | 70.0 | 0.0 | 582.0 | 0.0 | 20.0 | 1.0 | 263358.03 | 1.83 | 134.0 | 1.0 |
| 9 | 61.0 | 0.0 | 248.0 | 0.0 | 30.0 | 1.0 | 267000.0 | 0.7 | 136.0 | 1.0 |
| 10 | 40.0 | 0.0 | 478.0 | 1.0 | 30.0 | 0.0 | 303000.0 | 0.9 | 136.0 | 1.0 |

## B.6 Audit Trace: Fold 6 | Gen: 100%

| # | age | anaemia | creatinine_phosphoki | diabetes | ejection_fractio | high_blood_pressu | platelets | serum_creatinine | serum_sodium | sex |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 56.3215247395014 5 | 0.0 | 87.0 | 0.0 | 41.0 | 0.0 | 206033.786577636 23 | 1.32729868938971 63 | 134.0 | 1.0 |
| 2 | 56.0751996953196 25 | 0.0 | 83.0 | 0.0 | 26.0 | 1.0 | 235764.783906579 72 | 1.05668037707568 47 | 136.0 | 0.0 |
| 3 | 48.0825470398146 7 | 0.0 | 165.0 | 0.0 | 65.0 | 0.0 | 290364.783538498 9 | 0.73752193601941 81 | 138.0 | 1.0 |
| 4 | 74.6442053803757 4 | 0.0 | 48.0 | 0.0 | 30.0 | 0.0 | 261824.872094001 24 | 0.79923519389583 14 | 139.0 | 1.0 |
| 5 | 59.6973607742547 1 | 0.0 | 105.0 | 0.0 | 42.0 | 0.0 | 180102.374510114 02 | 0.90770846309304 86 | 140.0 | 1.0 |
| 6 | 69.0937702653246 2 | 1.0 | 10.0 | 0.0 | 33.0 | 0.0 | 196023.574508802 27 | 0.94579152449901 56 | 139.0 | 1.0 |
| 7 | 81.0 | 0.0 | 4540.0 | 0.0 | 35.0 | 0.0 | 231000.0 | 1.18 | 137.0 | 1.0 |
| 8 | 70.0 | 0.0 | 2695.0 | 1.0 | 40.0 | 0.0 | 241000.0 | 1.0 | 137.0 | 1.0 |
| 9 | 70.0 | 0.0 | 835.0 | 0.0 | 35.0 | 1.0 | 305000.0 | 0.8 | 133.0 | 0.0 |
| 10 | 80.0 | 0.0 | 776.0 | 1.0 | 38.0 | 1.0 | 192000.0 | 1.3 | 135.0 | 0.0 |

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## B.7 Audit Trace: Fold 7 | Gen: 100%

| # | age | anaemia | creatinine_phosphoki | diabetes | ejection_fractio | high_blood_pressu | platelets | serum_creatinine | serum_sodium | sex |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 56.704512797518916 | 0.0 | 214.0 | 0.0 | 41.0 | 0.0 | 230367.48729427822 | 0.7177098394469534 | 136.0 | 1.0 |
| 2 | 54.566768638113544 | 0.0 | 695.0 | 0.0 | 21.0 | 0.0 | 284647.8136160623 | 1.2489273190937667 | 132.0 | 1.0 |
| 3 | 63.298341704152233 | 1.0 | 249.0 | 0.0 | 31.0 | 1.0 | 336908.26769171166 | 0.941864248233696 | 141.0 | 1.0 |
| 4 | 69.753539881283599 | 1.0 | 23.0 | 0.0 | 48.0 | 0.0 | 315587.2408084735 | 1.0784230174558687 | 137.0 | 1.0 |
| 5 | 46.5698346099033 | 0.0 | 259.0 | 0.0 | 32.0 | 0.0 | 189792.44319504 | 1.3710316159212945 | 138.0 | 0.0 |
| 6 | 56.052428104345844 | 0.0 | 197.0 | 0.0 | 41.0 | 0.0 | 434700.6845417822525 | 1.2151852409417914 | 135.0 | 1.0 |
| 7 | 75.0 | 0.0 | 582.0 | 1.0 | 30.0 | 1.0 | 263358.03 | 1.83 | 134.0 | 0.0 |
| 8 | 53.0 | 0.0 | 207.0 | 1.0 | 40.0 | 0.0 | 223000.0 | 1.2 | 130.0 | 0.0 |
| 9 | 60.0 | 1.0 | 2281.0 | 1.0 | 40.0 | 0.0 | 283000.0 | 1.0 | 141.0 | 0.0 |
| 10 | 59.0 | 1.0 | 280.0 | 1.0 | 25.0 | 1.0 | 302000.0 | 1.0 | 141.0 | 0.0 |

## B.8 Audit Trace: Fold 8 | Gen: 100%

| # | age | anaemia | creatinine_phosphoki | diabetes | ejection_fractio | high_blood_pressu | platelets | serum_creatinine | serum_sodium | sex |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 55.852652489938554 | 0.0 | 118.0 | 0.0 | 22.0 | 0.0 | 317090.76533013675 | 1.0925465376809544 | 135.0 | 1.0 |
| 2 | 67.4285201749408 | 1.0 | -35.0 | 0.0 | 24.0 | 0.0 | 342059.86676234147 | 1.2553873335326595 | 140.0 | 1.0 |
| 3 | 55.448602510650915 | 0.0 | 52.0 | 1.0 | 31.0 | 1.0 | 334199.2835150518 | 0.9705934608070019 | 137.0 | 0.0 |
| 4 | 59.327824132554551 | 0.0 | 279.0 | 0.0 | 39.0 | 0.0 | 307651.19714522053 | 1.2699426265130636 | 137.0 | 1.0 |
| 5 | 81.393231022380099 | 0.0 | 258.0 | 0.0 | 38.0 | 0.0 | 246724.1303037444 | 1.2551396280528156 | 135.0 | 1.0 |
| 6 | 69.8762319824724 | 0.0 | 123.0 | 0.0 | 58.0 | 0.0 | 262096.3490297598 | 1.0624325487428776 | 137.0 | 1.0 |
| 7 | 65.0 | 0.0 | 224.0 | 1.0 | 50.0 | 0.0 | 149000.0 | 1.3 | 137.0 | 1.0 |
| 8 | 80.0 | 0.0 | 898.0 | 0.0 | 25.0 | 0.0 | 149000.0 | 1.1 | 144.0 | 1.0 |
| 9 | 66.0 | 1.0 | 72.0 | 0.0 | 40.0 | 1.0 | 242000.0 | 1.2 | 134.0 | 1.0 |
| 10 | 75.0 | 1.0 | 582.0 | 0.0 | 30.0 | 0.0 | 225000.0 | 1.83 | 134.0 | 1.0 |

# TabuLLM: AI Benchmark Report

## B.9 Audit Trace: Fold 9 | Gen: 100%

| # | age | anaemia | creatinine_phosphoki | diabetes | ejection_fractio | high_blood_pressu | platelets | serum_creatinine | serum_sodium | sex |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 58.5307447932723 | 0.0 | 75.0 | 1.0 | 23.0 | 0.0 | 251970.3395032108 | 1.3015430252397646 | 139.0 | 1.0 |
| 2 | 66.8095948190849 | 0.0 | 607.0 | 0.0 | 39.0 | 0.0 | 218998.61126255593 | 1.3060419817994287 | 139.0 | 1.0 |
| 3 | 54.827060690290004 | 0.0 | 197.0 | 0.0 | 29.0 | 0.0 | 206348.37328417954 | 0.5654969276413488 | 139.0 | 1.0 |
| 4 | 50.59846008731023 | 0.0 | 520.0 | 1.0 | 29.0 | 0.0 | 231986.22065471136 | 1.1060103199317273 | 143.0 | 1.0 |
| 5 | 58.953601415930976 | 0.0 | 182.0 | 1.0 | 21.0 | 0.0 | 299992.2227934919 | 1.2254498900047948 | 135.0 | 1.0 |
| 6 | 70.63549379122901 | 1.0 | 307.0 | 0.0 | 32.0 | 1.0 | 211826.66520963408 | 0.6197405063387407 | 138.0 | 1.0 |
| 7 | 55.0 | 0.0 | 572.0 | 1.0 | 35.0 | 0.0 | 231000.0 | 0.8 | 143.0 | 0.0 |
| 8 | 60.0 | 0.0 | 68.0 | 0.0 | 20.0 | 0.0 | 119000.0 | 2.9 | 127.0 | 1.0 |
| 9 | 70.0 | 0.0 | 97.0 | 0.0 | 60.0 | 1.0 | 220000.0 | 0.9 | 138.0 | 1.0 |
| 10 | 53.0 | 1.0 | 1808.0 | 0.0 | 60.0 | 1.0 | 249000.0 | 0.7 | 138.0 | 1.0 |

## B.10 Audit Trace: Fold 10 | Gen: 100%

| # | age | anaemia | creatinine_phosphoki | diabetes | ejection_fractio | high_blood_pressu | platelets | serum_creatinine | serum_sodium | sex |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 63.171668470411177 | 0.0 | 250.0 | 0.0 | 36.0 | 0.0 | 293813.38825279265 | 1.1992865801170893 | 135.0 | 1.0 |
| 2 | 66.918241397237004 | 1.0 | 203.0 | 0.0 | 50.0 | 0.0 | 280932.54232042655 | 1.1647858163875873 | 134.0 | 1.0 |
| 3 | 65.624693919361709 | 1.0 | 66.0 | 0.0 | 39.0 | 0.0 | 202920.30167439516 | 1.0365907693689702 | 144.0 | 1.0 |
| 4 | 46.397330478987 | 0.0 | 87.0 | 0.0 | 28.0 | 0.0 | 257919.9902670545 | 0.9705273684077791 | 139.0 | 1.0 |
| 5 | 76.940154930138001 | 0.0 | 166.0 | 0.0 | 29.0 | 0.0 | 288498.3981869848 | 1.2549437805465846 | 135.0 | 1.0 |
| 6 | 61.003689866560603 | 0.0 | 177.0 | 1.0 | 26.0 | 0.0 | 262426.0126219602 | 1.1187590257671347 | 132.0 | 1.0 |
| 7 | 60.0 | 0.0 | 897.0 | 1.0 | 45.0 | 0.0 | 297000.0 | 1.0 | 133.0 | 1.0 |
| 8 | 75.0 | 0.0 | 582.0 | 0.0 | 20.0 | 1.0 | 265000.0 | 1.9 | 130.0 | 1.0 |
| 9 | 45.0 | 1.0 | 1876.0 | 1.0 | 35.0 | 0.0 | 226000.0 | 0.9 | 138.0 | 1.0 |
| 10 | 75.0 | 1.0 | 246.0 | 0.0 | 15.0 | 0.0 | 127000.0 | 1.2 | 137.0 | 1.0 |